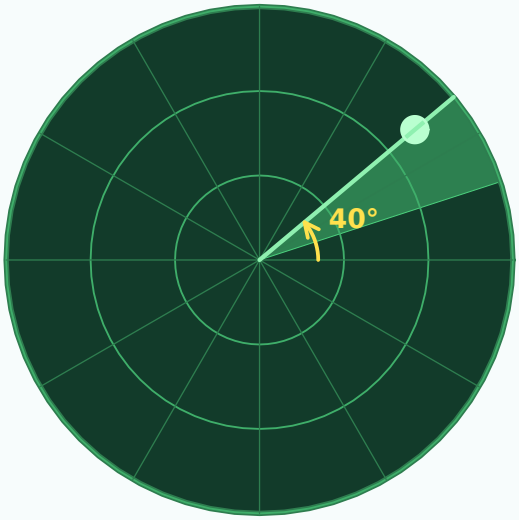
Chapter 33

Polar coordinates

turn, then walk



The radar operator doesn't say "3 km east, 2.5 km north". She says "40° round from east, 4 km out": turn to face the angle, then walk the distance. (Maths angles go anticlockwise from east. A navigator's bearing goes clockwise from north.)

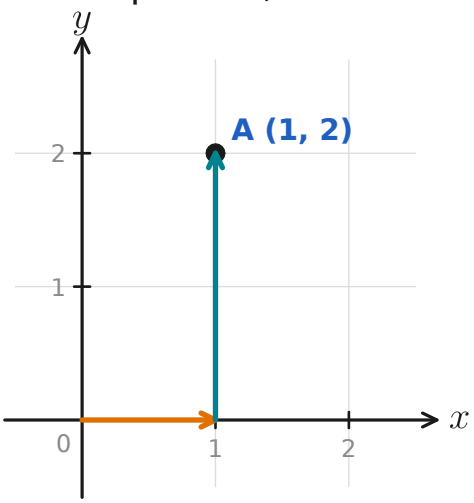
You've always said where things are with (x, y): so far across, so far up. **Polar coordinates** use a different pair: **how far from the centre** (r) and **which direction** (θ). Anything that spins or radiates (wheels, radar, waves, sound from a speaker) is easier this way.

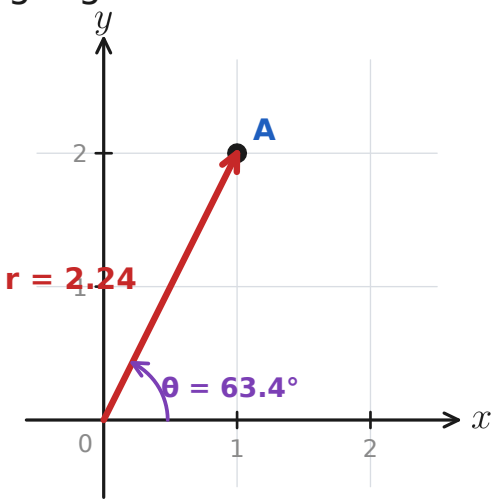
You'll translate between the two languages, dodge the calculator's blind spot, and draw some curves that would be horrible in x and y.

Goes with **Lecture 11** (Cartesian vs polar, converting, degrees vs radians, finding θ in each quadrant, polar curves, converting equations, the extended trig list, the cardioid).

Two ways to say where

The same point A, described in both languages:





Cartesian: walk 1 east, then 2 north

Polar: turn to 63.4°, then walk 2.24

Cartesian writes A as $(x, y) = (1, 2)$. Polar writes it as $(r, \theta) = (2.24, 63.4°)$. r is the **radius** (distance from the centre, the origin) and θ is the angle turned **anticlockwise from the positive x axis** (Chapter 27's convention).

Lecture 11: Cartesian is what graphs, slopes and vectors are built on; polar is how you describe circular motion and navigate on a round planet.

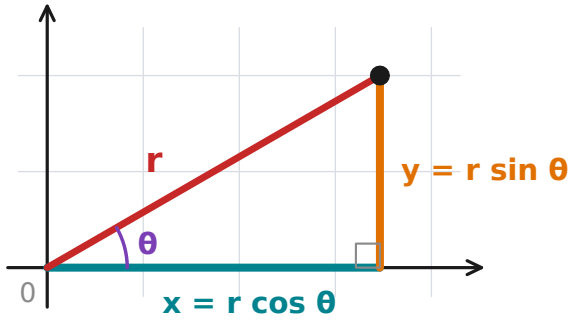
Polar to Cartesian: the triangle hiding inside

RULE: polar to Cartesian

$x = r\cos\theta$ and $y = r\sin\theta$.

This works for **any** angle: the signs of $\cos\theta$ and $\sin\theta$ put the point in the right quadrant automatically. No fix needed in this direction.

Drop a line from the point straight down to the x axis. You get a right-angled triangle with hypotenuse r and angle θ at the origin, so SOHCAHTOA (Chapter 28) does the rest:



$$x = r\cos\theta$$

CAH: adjacent = hyp × cos

$$y = r\sin\theta$$

SOH: opposite = hyp × sin

Example: a ship at $(r, \theta) = (4, 30°)$.

$$x = 4\cos 30° = 4 \times 0.866 = 3.46$$

$$y = 4\sin 30° = 4 \times 0.5 = 2$$

$$(r, \theta) = (4, 30°) \longrightarrow (x, y) = (3.46,\ 2)$$

Slide slip: Lecture 11's summary slide writes $y = x\sin\theta$. It should be $y = r\sin\theta$ (the hypotenuse is r, not x).

Cartesian to polar, and the calculator's blind spot

RULE: Cartesian to polar

$r = \sqrt{x^2 + y^2}$ (Pythagoras; r is never negative here).

$\theta = \tan^{-1}\dfrac{y}{x}$ **plus a correction** that depends on the quadrant: the calculator only answers between $-90°$ and $90°$.

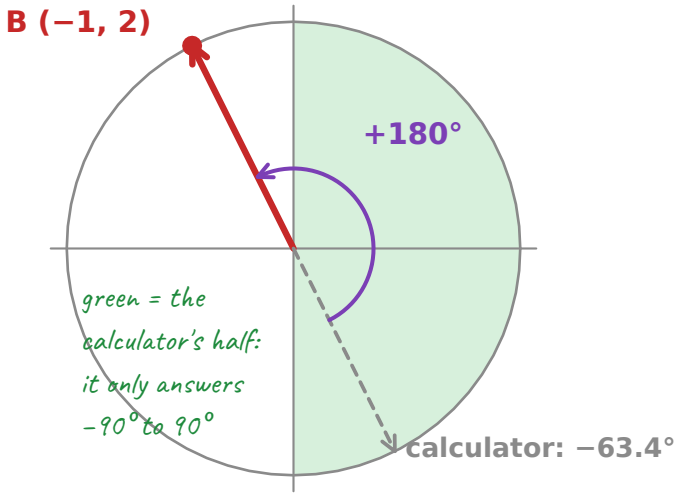
Always sketch the point first so you know which quadrant the answer must be in.

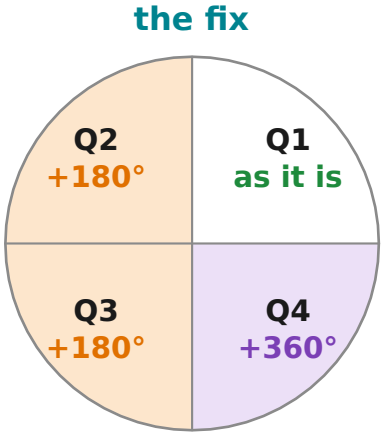
Going the other way, Pythagoras gives the distance and TOA gives the angle:

$$r = \sqrt{x^2 + y^2}, \qquad \theta = \tan^{-1}\frac{y}{x}$$

Lecture 11: $A = (1, 2)$ in quadrant 1. $r = \sqrt{1+4} = \sqrt{5} = 2.24$, $\theta = \tan^{-1}\frac{2}{1} = 63.43°$ (1.107 rad). Perfect.

Now $B = (-1, 2)$, in quadrant 2. The calculator says $\tan^{-1}\frac{2}{-1} = \tan^{-1}(-2) = -63.43°$. That points down into quadrant 4: **the wrong way!**





Why +180°? The calculator can't tell $\frac{2}{-1}$ from $\frac{-2}{1}$: both are -2. So it always answers in the right-hand half. If the point is really on the left (quadrants 2 or 3), its direction is exactly opposite: turn another half-turn.

$$\theta_B = -63.43^\circ + 180^\circ = 116.57^\circ \quad (-1.107 + \pi = 2.03 \text{ rad})$$

The quadrant fix in full. Work out $\tan^{-1}\frac{y}{x}$ on the calculator, then:

Point is in	Signs	Add (degrees)	Add (radians)	θ ends up in
Quadrant 1	x > 0, y > 0	nothing	nothing	0° to 90° (0 to π/2)
Quadrant 2	x < 0, y > 0	180°	π	90° to 180° (π/2 to π)
Quadrant 3	x < 0, y < 0	180°	π	180° to 270° (π to 3π/2)
Quadrant 4	x > 0, y < 0	360°	2π	270° to 360° (3π/2 to 2π)

RULE: points on the axes

No calculator needed (and $\tan^{-1}\frac{y}{0}$ gives an error anyway). Read the angle off the sketch:
positive x axis 0; positive y axis $90^\circ = \frac{\pi}{2}$; negative x axis $180^\circ = \pi$; negative y axis $270^\circ = \frac{3\pi}{2}$. And r is just the distance from O.

$$(0, -3): \quad r = 3, \quad \theta = 270^\circ = \frac{3\pi}{2}$$ ← straight down

$$(-4, 0): \quad r = 4, \quad \tan^{-1}\frac{0}{-4} = 0^\circ \to 0^\circ + 180^\circ = 180^\circ = \pi$$

x < 0: the fix still works

Lecture 11's four test points. Watch the calculator column repeat itself:

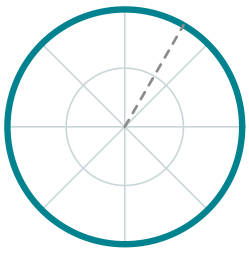
point	y ÷ x	calculator	quadrant fix	true angle
(1, 1)	1	45°	Q1: as it is	**45° = π/4**
(−1, 1)	−1	−45°	Q2: +180°	**135° = 3π/4**
(−1, −1)	1	45°	Q3: +180°	**225° = 5π/4**
(1, −1)	−1	−45°	Q4: +360°	**315° = 7π/4**

Slide slip: Lecture 11 labels 225° as $\frac{4\pi}{3}$; it's $\frac{5\pi}{4}$ ($\frac{225}{180} = \frac{5}{4}$). $\frac{4\pi}{3}$ would be 240°.

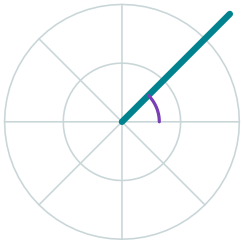
Degrees or radians? Lecture 11: **calculate in radians**, and present in degrees only when the context wants it (navigation). If a question names a format, use it; if not, either is fine. Check the calculator's DEG/RAD mode first (Chapter 27).

Drawing with a spinning pen

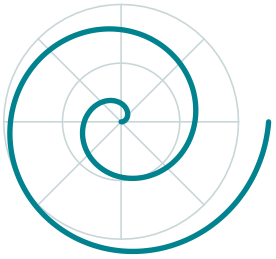
Picture a pen on a spinning arm, like the radar sweep. As the arm turns through θ, you slide the pen in or out to distance r. A **polar equation** is the rule that says how far out the pen is at each angle.



r = 2, any θ:
a circle

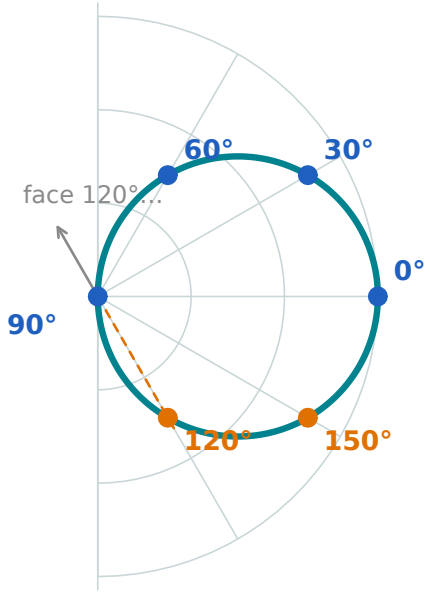


θ = 45°, any r:
a ray from the centre



r grows with θ
(r = 0.2θ): a spiral

Lecture 11's example: $r = 1.5\cos\theta$. Make a table, then plot each point by turning to θ and walking r:



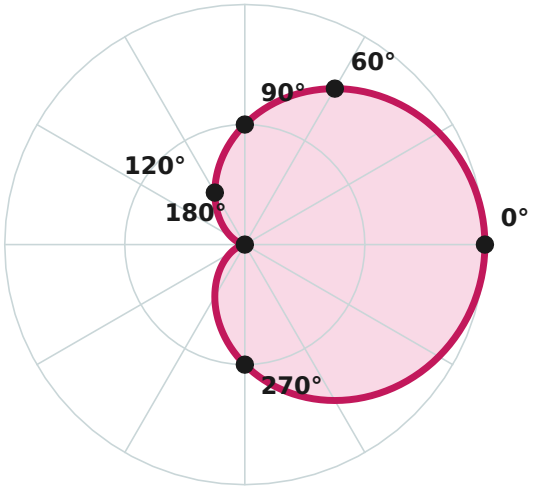
θ	cos θ	r = 1.5 cos θ
0°	1.00	**1.50**
30°	0.87	**1.30**
60°	0.50	**0.75**
90°	0.00	**0.00**
120°	−0.50	**−0.75**
150°	−0.87	**−1.30**
180°	−1.00	**−1.50**

Negative r: face θ, then walk BACKWARDS. So 120° lands where 300° would: the pen goes round the same circle again.

Surprise: it's a **circle** of diameter 1.5 sitting on the origin. In x and y that's $(x - 0.75)^2 + y^2 = 0.75^2$, which is much uglier to plot.

The cardioid: a heart from one line of maths

Change the rule to $r = 1 + \cos\theta$. The pen starts 2 out, comes in to 1 when facing up, and reaches 0 (the centre) facing left:



θ = 0°: r = 1 + 1 = 2 (far right)
θ = 90°: r = 1 + 0 = 1 (straight up)
θ = 120°: r = 1 − 0.5 = 0.5
θ = 180°: r = 1 − 1 = 0 (the dent)
θ = 270°: r = 1 (straight down)

"Cardioid" means heart-shaped. Microphones that hear mostly from the front have this pick-up pattern, and it turns up in roller and cam design (Lecture 11).

Converting equations: swap x and y out

RULE: converting equations, both ways

Cartesian → polar: replace x by $r\cos\theta$ and y by $r\sin\theta$ (and $x^2 + y^2$ by r^2), then make r the subject.

Polar → Cartesian: look for $r\cos\theta$, $r\sin\theta$ and r^2 and swap them for x, y and $x^2 + y^2$. If there is a lone $\cos\theta$ or $\sin\theta$, **multiply both sides by r** first.

A fixed angle $\theta = \alpha$ is the line through O: $\dfrac{y}{x} = \tan\alpha$.

To turn a Cartesian equation into a polar one: **replace every x by $r\cos\theta$ and every y by $r\sin\theta$**, collect the r terms, then make r the subject (Chapter 14).

Lecture 11: the straight line $y = 2x + 2$.

$$r\sin\theta = 2r\cos\theta + 2 \quad \longleftarrow \text{swap y and x}$$

$$r\sin\theta - 2r\cos\theta = 2 \quad \longleftarrow \text{r terms together}$$

$$r(\sin\theta - 2\cos\theta) = 2 \quad \longleftarrow \text{factor out r (Chapter 12)}$$

$$r = \frac{2}{\sin\theta - 2\cos\theta}$$ ← ÷ the bracket

Check one point: at $\theta = 90°$, $r = \frac{2}{1-0} = 2$: straight up 2, the point $(0, 2)$. And $y = 2x + 2$ does cross the y axis at 2. ✓

Lecture 11: the oval $x^2 + 2y^2 = 9$.

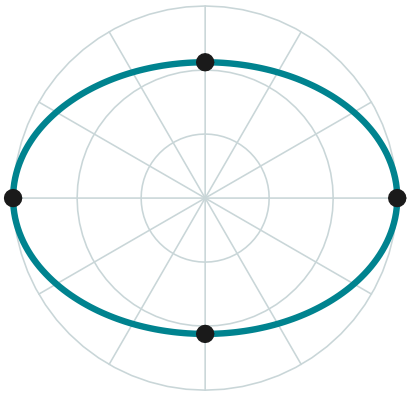
$$(r\cos\theta)^2 + 2(r\sin\theta)^2 = 9$$ ← swap

$$r^2(\cos^2\theta + 2\sin^2\theta) = 9$$ ← factor out r^2

$$r^2(\underbrace{\cos^2\theta + \sin^2\theta}_{=1} + \sin^2\theta) = 9$$ ← split 2 $\sin^2$ into $\sin^2 + \sin^2$, then Chapter 30's identity

$$r^2 = \frac{9}{1 + \sin^2\theta} \quad \Rightarrow \quad r = \frac{3}{\sqrt{1 + \sin^2\theta}}$$

Check: $\theta = 0$ gives $r = 3$, the point $(3, 0)$: $9 + 0 = 9$ ✓. $\theta = 90°$ gives $r = \frac{3}{\sqrt{2}} = 2.12$, the point $(0, 2.12)$: $2 \times 2.12^2 = 9$ ✓.



Lecture 11's "Polar curves 3" slide is this oval. As the pen turns, r changes smoothly: largest (r = 3) facing 0 and π, smallest (r = 2.12) facing π/2 and 3π/2. So the oval is wider than it is tall.

Converting back, and quick wins

Circle centred on O: $x^2 + y^2 = 9$.

$$r^2 = 9 \;\Rightarrow\; r = 3$$ ← $x^2 + y^2$ is r^2

Line through O: $y = x$.

$$r\sin\theta = r\cos\theta \;\Rightarrow\; \tan\theta = 1 \;\Rightarrow\; \theta = \tfrac{\pi}{4}$$ ← ÷ r cos θ

(With negative r allowed, $\theta = \frac{\pi}{4}$ covers the whole line, including the part in quadrant 3.)

Polar → Cartesian: $r = 4\cos\theta$. There is a lone $\cos\theta$, so multiply by r:

$$r^2 = 4r\cos\theta$$ ← × r

$$x^2 + y^2 = 4x$$ ← $r^2 \to x^2 + y^2$, r cos θ → x

$$x^2 - 4x + y^2 = 0 \;\Rightarrow\; (x-2)^2 + y^2 = 4$$ ← complete the square

A circle of radius 2 centred at $(2, 0)$, passing through O, just like $r = 1.5\cos\theta$ earlier.

✗ Common mistakes

✗ $r = 4\cos\theta \;\to\; r = 4x$ — x is r cos θ, not cos θ: multiply by r first

✓ $r^2 = 4r\cos\theta = 4x$

✗ $x^2 + y^2 = 9 \;\to\; r = 9$ — $x^2 + y^2$ is r^2, not r

✓ $r^2 = 9,\; r = 3$

Lecture 11's extended trig list

Converting equations often needs these. The first four are the compound angle rules (Chapter 30 met the first). Notice the sign **flips** in the cosine ones:

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

Now just **set** $B = A$ in the "+" versions, and the **double angle** rules drop out:

$$\sin 2A = 2 \sin A \cos A \longleftarrow \sin A \cos A + \cos A \sin A$$

$$\cos 2A = \cos^2 A - \sin^2 A = 2\cos^2 A - 1 = 1 - 2\sin^2 A$$

The last two come from swapping $\sin^2 A = 1 - \cos^2 A$ (or the other way round) into the first one. Quick test with $A = 30°$: $\sin 60° = 0.866$ and $2 \times 0.5 \times 0.866 = 0.866$ ✓.

Rules in this chapter

Learn these. Every worked example in the chapter is one of these rules applied step by step.

Polar → Cartesian	$x = r\cos\theta, \qquad y = r\sin\theta$ works for any angle; no fix needed
Distance	$r = \sqrt{x^2 + y^2}$
Angle	$\theta = \tan^{-1}\dfrac{y}{x}$ then the quadrant fix
Quadrant fix	Q1: nothing; Q2 and Q3: + 180° (+ π); Q4: + 360° (+ 2π)
Points on the axes	read off the sketch: 0, 90° (π/2), 180° (π), 270° (3π/2)
Links	$x^2 + y^2 = r^2, \qquad \tan\theta = \dfrac{y}{x}$
Cartesian → polar equation	swap $x \to r\cos\theta$, $y \to r\sin\theta$, make r the subject
Polar → Cartesian equation	multiply by r if needed, then swap $r\cos\theta \to x$, $r\sin\theta \to y$, $r^2 \to x^2 + y^2$
Simple curves	$r = a$: circle radius a; $\theta = \alpha$: line through O; $r = a\cos\theta$: circle through O
Double angle	$\sin 2A = 2\sin A\cos A, \quad \cos 2A = \cos^2 A - \sin^2 A$

Your turn

Sketch the point or curve first. For θ from $\tan^{-1}$, always ask "which quadrant?". Answers in grey.

Polar to Cartesian

1. $(r, \theta) = (6, 60°)$
2. $(r, \theta) = (2, 3\pi/4)$

1. (3, 5.20) 2. (−1.41, 1.41)

Cartesian to polar (mind the quadrant!)

3. (3, 4)
4. (−3, 4)
5. (−2, −2)
6. (5, −5)

3. (5, 53.13°) 4. (5, 126.87°) 5. (2.83, 225°) 6. (7.07, 315°)

Curves and equations

7. Sketch $r = 3$. What shape is it?
8. Where is the pen for $r = 1 + \cos\theta$ at $\theta = 90°$? Give x and y.
9. Write the line $x = 4$ in polar form.
10. Write the circle $x^2 + y^2 = 25$ in polar form.

7. a circle of radius 3 8. r = 1, the point (0, 1) 9. $r\cos\theta = 4$, so $r = 4/\cos\theta$ 10. $r^2 = 25$, so r = 5

Double angle

11. Use $\sin 2A = 2\sin A\cos A$ to find sin 90° from A = 45°.

11. $2 \times 0.7071 \times 0.7071 = 1$

Find the unknown

12. Write $r = 6\sin\theta$ in Cartesian form and describe the curve.
13. Write the line $y = \sqrt{3}\,x$ in polar form.
14. Write the point (0, −5) in polar form, in degrees and radians.
15. A point has r = 8, y = 4 and is left of the y axis. Find θ and x.
16. A point has x = −3 and θ = 135°. Find y and r.

12. × r: $r^2 = 6r\sin\theta$, $x^2 + y^2 = 6y$, $x^2 + (y - 3)^2 = 9$: circle, centre (0, 3), radius 3 13. $\tan\theta = \sqrt{3}$, θ = π/3 (60°) 14. on the negative y axis: (5, 270°) = (5, 3π/2) 15. $\sin\theta = 4/8 = 0.5$ and x < 0, so θ = 150°; $x = 8\cos 150° \approx -6.93$ 16. $y = x\tan 135° = (-3)(-1) = 3$; $r = \sqrt{(9 + 9)} \approx 4.24$